# Vineeth Kirandumkara

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Embedded systems engineer looking for mid-level product development role.
5+ years of experience blending IoT with product development, skilled in C/C++ and Python, creating reliable, scalable, user-centered smart devices and connected solutions from concept to deployment.

## Work Experience

**Embedded Software Engineer**
Grenova
Richmond, VA, USA
Nov ‘23 – Aug ‘24

- Developed embedded C++ firmware for TipNovus 2.0 and designed the architecture for dependency management using CMake.
- Authored advanced memory pool structures to allow for more control over factory class memory allocation.
- Implemented command/reaction protocols and system diagnostics to allow for user-driven customization for washing and drying actions.

**Embedded Software Engineer**
Iontra Inc
Denver, CO, USA
Nov 2022 – Nov 2023

- Automated the firmware verification processes across 2000+ new boards; transitioned systems to manufacturing.
- Developed firmware for real-time monitoring of pouch cell metrics, for improved safety alerts.
- Modularized CMake project architecture, developed Zephyr RTOS-based task scheduling and added OS/HAL abstraction improvements.

**Embedded Systems Engineer**
Platform Aerospace
Hollywood, MD, USA
Feb 2021 – Sept 2023

- Refactored Vanilla Unmanned codebase with focus on improved memory management to complete world record-breaking UAV endurance flight.
- Delivered 50% reduction in dynamic memory and 40% faster processing cycle by leveraging C++ pattern optimizations.
- Developed Hardware-in-the-Loop simulator to reduce roadblocks in system testing due to resource limitations.

**Engineer 1**
Daimler Trucks North America
Portland, OR, USA
July 2019 – Feb 2021

- Designed and implemented a data reporting tool (Python + Alteryx) to streamline validation workflows and improve cross-team visibility into large-scale test results.
- Collaborated with Product Validation Engineering to create reporting methods for tested ECUs, supporting more informed product decision-making.

**Mobile App Dev Intern**
CapTech Ventures Inc
Richmond, VA, USA

- Partnered with stakeholders to design and deliver a native iOS app with AWS backend integration, enabling vehicle browsing through dynamic filtering.
- Coordinated iOS–backend development through REST API refactoring and rapid iteration cycles, including 24-hour bug resolution to meet

May 2018 – Aug 2018 critical demo deadlines.

## Work Related Projects

Hot Stone

- Developed an IoT-based device to remotely share warmth between two users using capacitive touch sensors and heating elements.

Gesture Recognizer

- Built a prototype wearable to recognize real-time signals from flex sensors as hand gestures using deep learning models trained with Edge Impulse.

Traveler

- Created a mobile app to track a user's locations of interests and notify them when nearby. Written in Flutter with a Supabase backend and Google Maps and Places API.

Lumos

- Engineered an IoT light-wall control system that enabled users to select specific LED nodes, and control hue and saturation via a magnetometer matrix and rotary selector.

## Education

University College London • 2024 – 2025

The Bartlett Faculty of the Built Environment

**MSc Connected Environments**

- The dissertation involved the development of a low-cost ultrasonic anemometer to measure wind speed in scaled urban canyon models, with the aim to democratize environmental sensing for urban design, pedestrian comfort, and evidence-based policymaking.

Virginia Tech • 2014 – 2019

Bradley Department of Electrical and Computer Engineering

**BS Computer Engineering**

## Interests

Architecture • Industrial Design • Golf • Padel • Football